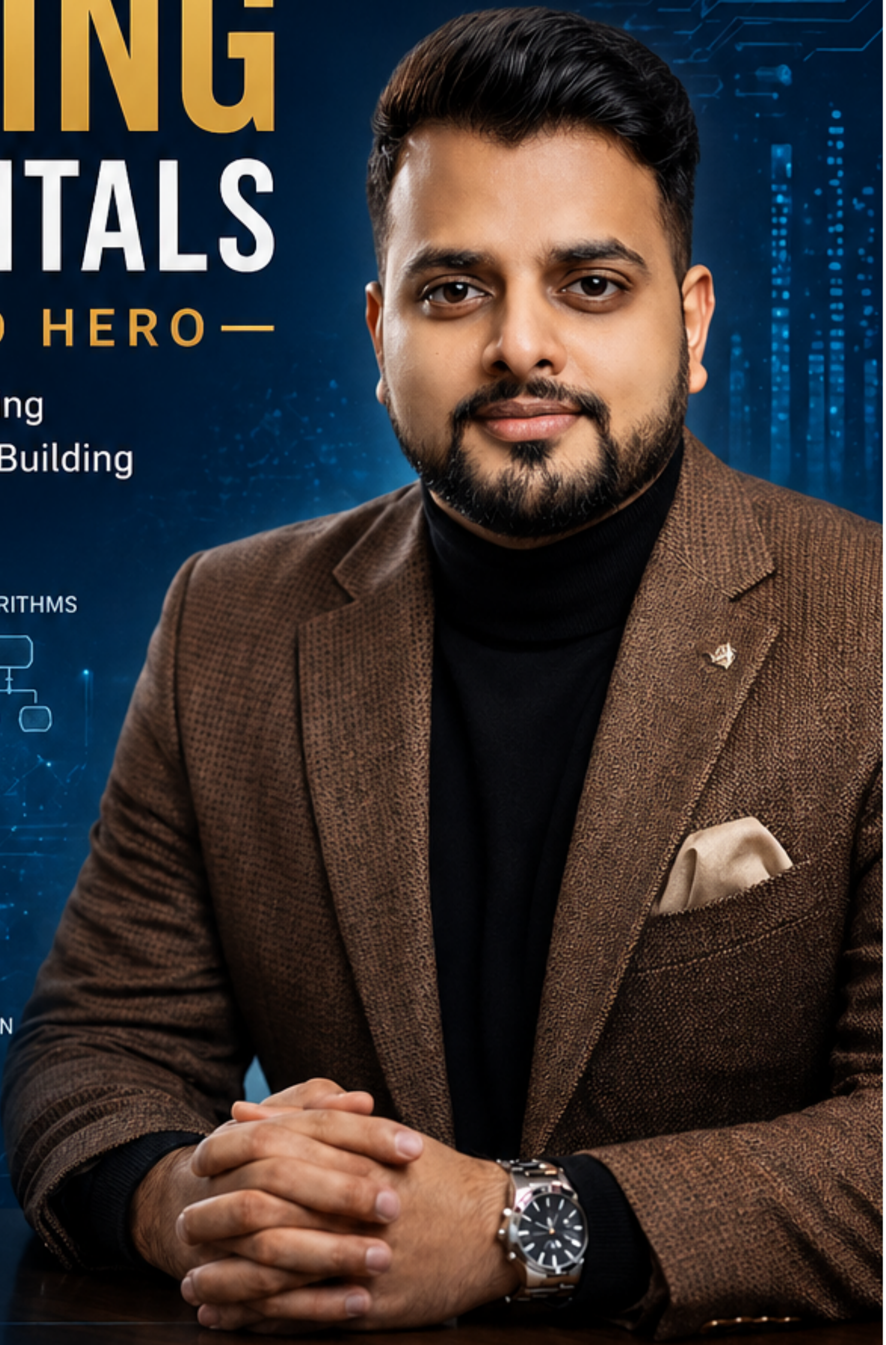
MACHINE LEARNING FUNDAMENTALS

FROM ZERO TO HERO

A Complete Guide to Understanding Machine Learning Concepts and Building Real-World AI Solutions



- Core Concepts
- Hands-on Examples
- Real-World Projects
- Industry Best Practices
- From Basics to Advanced

BY ABHISHEK PANDA

AI ARCHITECT | AZURE AI ARCHITECT

STEP-BY-STEP LEARNING

PRACTICAL IMPLEMENTATIONS

REAL-WORLD INSIGHTS

CAREER ACCELERATION

Machine Learning — Zero to Hero

Complete Master Syllabus (2026 Edition)

By **Abhishek Panda**

AI Architect | Azure AI Architect

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Module 0: Machine Learning Foundations

Introduction to Artificial Intelligence Ecosystem

- AI vs ML vs Deep Learning vs Generative AI
- Types of Artificial Intelligence
- Real-World Applications of Machine Learning
- Industry Use Cases Across Domains
- Machine Learning Lifecycle Overview

Core ML Concepts

- Supervised vs Unsupervised vs Reinforcement Learning
- Semi-Supervised Learning
- Online vs Offline Learning
- Batch vs Incremental Learning
- Parametric vs Non-Parametric Models

Fundamental Terminologies

- Features / Labels / Targets
- Training / Validation / Test Sets
- Epoch / Batch / Iteration
- Model / Algorithm / Parameters / Hyperparameters
- Bias / Variance
- Underfitting / Overfitting
- Curse of Dimensionality

Module 1: Mathematics & Statistics for Machine Learning

Linear Algebra

- Scalars, Vectors, Matrices, Tensors
- Matrix Operations
- Dot Product / Cross Product
- Matrix Transpose / Inverse

- Determinants
- Eigenvalues / Eigenvectors

Calculus

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- Partial Derivatives
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Probability

- Probability Fundamentals
- Conditional Probability
- Bayes Theorem
- Probability Distributions
  - Normal Distribution
  - Binomial Distribution
  - Poisson Distribution
  - Uniform Distribution

Statistics

- Mean / Median / Mode
- Variance / Standard Deviation
- Correlation / Covariance
- Hypothesis Testing
- p-value / Confidence Interval
- Statistical Significance

Module 2: Python & Data Science Stack for ML

Python for ML

- Python Refresher
- Functions / OOP / Modules
- Virtual Environments
- Package Management

Libraries

- **NumPy** Deep Dive
- **Pandas** Deep Dive
- **Matplotlib**
- **Seaborn**
- **Plotly**
- **SciPy**
- Statsmodels

Jupyter & IDE Workflow

- Jupyter Notebook Best Practices
- VS Code / PyCharm for ML

Module 3: Data Preprocessing & Feature Engineering

Data Cleaning

- Missing Values Handling
- Duplicate Removal
- Data Type Corrections
- Outlier Detection & Treatment

Feature Scaling

- Standardization
- Normalization
- Robust Scaling
- Min-Max Scaling

Encoding Techniques

- Label Encoding
- One-Hot Encoding
- Ordinal Encoding
- Target Encoding
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Feature Engineering

- Polynomial Features
- Binning / Bucketing
- Interaction Features
- Date/Time Feature Engineering
- Text Feature Extraction Basics

Feature Selection

- Filter Methods
- Wrapper Methods
- Embedded Methods
- Recursive Feature Elimination
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- Class Weights
- Random Oversampling / Undersampling
- SMOTE
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Module 4: Exploratory Data Analysis (EDA)

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- Univariate Analysis
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Visualization

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- Violin Plots
- Heatmaps
- Pairplots
- Correlation Matrices

Advanced EDA

- Distribution Analysis
- Skewness / Kurtosis
- Multicollinearity Detection
- Data Leakage Detection

Module 5: Supervised Learning — Regression

Linear Models

- Simple Linear Regression
- Multiple Linear Regression
- Polynomial Regression

Regularized Regression

- Ridge Regression
- Lasso Regression
- ElasticNet

Advanced Regression Models

- Decision Tree Regressor
- Random Forest Regressor
- Gradient Boosting Regressor
- XGBoost Regressor
- LightGBM Regressor
- CatBoost Regressor

Regression Metrics

- MAE
- MSE
- RMSE
- R^2 Score
- Adjusted R^2

Module 6: Supervised Learning — Classification

Linear Classification

- Logistic Regression
- Sigmoid Function
- Cost Function

Distance Based Models

- K-Nearest Neighbors

Probabilistic Models

- Naive Bayes
  - Gaussian
  - Multinomial
  - Bernoulli
  - Complement

Margin Based Models

- Support Vector Machines (SVM)
- Kernel Trick
- Non-Linear SVM

Tree Based Models

- Decision Tree Classifier
- Random Forest Classifier
- Extra Trees Classifier

Boosting Models

- AdaBoost
- Gradient Boosting
- XGBoost
- LightGBM
- CatBoost

Classification Metrics

- Accuracy
- Precision
- Recall
- F1 Score

- ROC-AUC
- PR-AUC
- Confusion Matrix

Module 7: Ensemble Learning & Optimization

Ensemble Concepts

- Bagging
- Boosting
- Stacking
- Blending

Hyperparameter Optimization

- Grid Search
- Random Search
- Bayesian Optimization
- Optuna Introduction

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- Time Series Split

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- K-Means++
- Hierarchical Clustering
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- DBSCAN
- OPTICS
- Gaussian Mixture Models
- Mean Shift
- Affinity Propagation
- Fuzzy C-Means

Dimensionality Reduction

- PCA
- Kernel PCA
- ICA
- NMF
- t-SNE
- UMAP
- Isomap
- LLE

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- FP-Growth
- ECLAT

Module 9: Time Series Machine Learning

Foundations

- Time Series Concepts
- Trend / Seasonality / Cyclicality
- Stationarity

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- SARIMA
- Prophet

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- Feature Engineering for Time Series
- XGBoost for Forecasting
- Rolling Window Validation

Module 10: Anomaly Detection

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- Isolation Forest
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Module 13: Explainable AI & Responsible AI

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Module 14: Model Deployment

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- FastAPI Deployment

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- Dockerizing ML Apps

- Docker Compose

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- Kubernetes Basics
- Model Serving Concepts

Module 15: Azure Machine Learning Platform

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- Azure ML Workspace
- Compute Instances / Clusters
- Datastores / Data Assets
- Environments / Components

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- Azure ML SDK v2
- Experiment Tracking
- Pipelines
- AutoML
- HyperDrive / Sweeps

Deployment

- Managed Online Endpoints
- Batch Endpoints
- AKS Deployment
- Azure Container Apps

Governance

- Model Registry
- Responsible AI Dashboard
- Monitoring with Azure Monitor

Module 16: MLOps & Production Engineering

MLOps Fundamentals

- CI/CD for ML
- ML Pipelines
- Model Registry

Experiment Tracking

- MLflow
- Weights & Biases

Data & Model Versioning

- DVC
- Git Integration

Monitoring

- Drift Detection
- Performance Monitoring
- Data Quality Monitoring

Production Strategies

- Canary Deployment
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- Shadow Deployment
- Retraining Pipelines

Module 17: Enterprise Capstone Projects

Beginner Projects

- House Price Prediction
- Titanic Survival Prediction

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- Customer Churn Prediction
- Fraud Detection
- Recommendation System

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- Demand Forecasting Pipeline
- Predictive Maintenance System
- Azure End-to-End Production ML System
- Enterprise MLOps Pipeline Project

Bonus Module: Career & Interview Preparation

- ML Interview Questions
- System Design for ML Engineers
- Portfolio Building
- Resume Optimization
- GitHub Project Structuring
- Azure ML Certification Guidance

Recommended Core Libraries / Tools Coverage

1. Python
2. NumPy
3. Pandas
4. Matplotlib
5. Seaborn
6. Plotly
7. Scikit-Learn
8. XGBoost
9. LightGBM
10. CatBoost
11. Optuna
12. MLflow
13. DVC
14. FastAPI
15. Docker
16. Kubernetes
17. Azure ML SDK
18. Azure CLI
19. GitHub Actions
20. Terraform/Bicep (Optional)

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Theory → Math → Hands-on Code → Visualisation → Interview Questions → Industry Use Case → Azure Production Implementation

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